

Course: Transversal ISI Project (ING4187)

Project 2: Kingdom Communication Solutions (KCS) Media Studio - Web-Based AI Video Transcription and Multilingual Caption Editing

Authors:

Abba Adji Wriwiya (2324i177)
Donfack Ateufack Robine Ninnone (2324i272)
Djiofack Nguimgo Natacha Maurelle (2324i188)
Nguisseu Assonzap Thomas Karim (2324i148)
Simo Noumsi Abraham (2324i242)
Tagne Yombo Manuel Fadhil (2324i319)
Tsafack Tene Thierry Emmanuel (2324i171)

Supervisor:

Dr.-Ing. Philippe Tamla

Professor for Data Science and Artificial Intelligence
PostDoc Researcher, Department of Multimedia and Internet Applications
Faculty of Mathematics and Computer Science
Universitätsstr. 1, 58097 Hagen

September 25, 2026

1 Chapter 1: Introduction

1.1 General Background and Trends

In the contemporary digital era, multimedia communication has become a predominant vehicle for global knowledge dissemination and education [1]. Advances in artificial intelligence, large language models, and self-supervised neural speech representations have catalyzed automated video localization pipelines capable of transcribing, subtitling, and translating audiovisual content at scale [2, 3]. Modern architectures increasingly embed multimodal workflows that automatically align phonemes and perform cross-lingual representation learning across widespread languages [4–6]. However, these automated workflows overwhelmingly serve commercially dominant languages, relying on massive supervised corpora that remain scarce for the vast majority of indigenous African tongues. In multilingual sub-Saharan regions, the absence of tailored digital authoring tools disenfranchises non-Western linguistic communities, reinforcing an acute communicative divide where current technologies fail to support tonal variations, complex orthographies, and regional dialects [7–9].

1.2 Motivation

The Republic of Cameroon represents a highly dense linguistic ecosystem comprising over two hundred and fifty distinct mother tongues, including Ghɔmálá', Ewondo, and Basaa. Despite this cultural richness, local populations face systemic exclusion from digital multimedia due to the lack of computational resources and standardized digital corpora for these languages [10]. Two primary motivations guide this research. **MS1:** There is a critical need for reliable, validated speech and text resources for low-resource Cameroonian languages, because the lack of standardized corpora severely limits the development of effective automatic speech recognition and multimedia localization technologies, thereby threatening the preservation of the country's intangible linguistic heritage [11, 12]. **MS2:** Enhancing socioeconomic and civic inclusivity requires assistive video editing technologies, yet existing web-based transcription and caption-editing environments are not sufficiently adapted to the regional linguistic variation, diacritics, and specific orthographic requirements of low-resource Cameroonian languages.

1.3 Overall Goal

The overall goal of this research is to design, develop, and evaluate a web-based AI-assisted video transcription and multilingual caption-editing platform, known as KCS Media Studio, that natively supports Cameroonian languages while integrating validated linguistic resources and adaptive mechanisms for handling regional language variation.

1.4 Problem Statements

Achieving this goal requires overcoming significant technical hurdles highlighted by two core problem statements. **PS1:** Existing speech recognition and transcription systems are fundamentally constrained by the limited availability of standardized, validated speech datasets, phonological dictionaries, and acoustic models for tonal Cameroonian mother tongues, rendering mainstream algorithms prone to high error rates. **PS2:** Conventional video transcription and caption-editing environments do not adequately accommodate regional accents, slang, dialectal variation, and timeline desynchronization issues inherent to the General Alphabet of Cameroonian Languages, forcing media creators into labor-intensive manual timeline adjustments that fail to reflect the cultural nuance of native speech [13].

1.5 Research Questions

To systematically address these sociolinguistic and technical barriers, this study investigates two central research questions. **RQ1:** How can the editing environment accommodate variations in regional accents or slang without alienating users from different villages or communities who speak the same base language? **RQ2:** How can a multilingual speech-transcription tool be developed so that users can smoothly edit videos while integrating written lexical resources and voice recordings for low-resource Cameroonian languages seamlessly within a browser application?

1.6 Nunamaker Methodology

This research formally adopts the multi-methodological information systems development framework established by Nunamaker et al. [14], which bridges theoretical modeling and empirical observation through four phases: Observation, Theory Building, Implementation, and Evaluation. Rather than merely following a generalized software lifecycle, this framework is intimately linked to the KCS Media Studio project by ensuring that every conceptual algorithm for handling tonal variations directly translates into an engineered platform feature, which is then strictly evaluated by native speakers.

1.7 Research Objectives

Following the Nunamaker framework, the research questions are decomposed into specific Research Objectives (**ROs**). For **RQ1**, the objectives are defined as **O1.1.O (Observation)** to investigate existing transcription systems regarding dialectal and orthographic handling; **O1.2.T (Theory Building)** to develop a conceptual design for an adaptive editing environment incorporating localized Cameroonian vocabulary; **O1.3.I (Implementation)** to implement these adaptive caption functionalities within the KCS Media Studio prototype; and **O1.4.E (Evaluation)** to evaluate whether the environment effectively supports regional variation for the intended indigenous users. For **RQ2**, the objectives are structured as **O2.1.O (Observation)** to examine multilingual speech approaches and requirements for integrating lexical data into web workflows; **O2.2.T (Theory Building)** to design an architecture connecting audio recordings, lexicons, and video caption editing; **O2.3.I (Implementation)** to implement the proposed transcription and resource integration components; and **O2.4.E (Evaluation)** to assess transcription performance, integration reliability, and overall user experience through controlled trials.

1.8 Approach Section

The methodological approach begins with community-based corpus acquisition using structured mobile forms to compile aligned audio pairs and core vocabulary across the target pilot languages [15]. These corpora directly inform the fine-tuning of self-supervised multilingual representations. The software artifact is then engineered as a cloud-native platform featuring an automated rendering backend and an intuitive frontend, which is ultimately validated through controlled laboratory trials with native translators measuring throughput against traditional baselines [16].

1.9 Outline Section

The remainder of this work is structured as follows. Chapter 2 presents the State of the Art regarding low-resource speech technologies and digital linguistic resources. Chapter 3 outlines the system design and architecture of the platform. Chapter 4 details the proof-of-concept implementation. Chapter 5 presents the evaluation methodology alongside the experimental results. Finally, Chapter 6 concludes the study by summarizing the main contributions and outlining potential avenues for future work.

1.10 Structure of Chapter 2 (State of the Art)

To systematically establish the foundation for this study, the State of the Art (Chapter 2) is structured into three primary domains. First, it reviews current automatic speech recognition architectures for African languages. Second, it analyzes existing web-based multimedia transcription platforms and their limitations regarding low-resource localization. Finally, it examines the availability of digital linguistic corpora for Cameroonian dialects to explicitly highlight the technological gaps that this research aims to resolve.

References

[1] R. E. Mayer. *Multimedia Learning*. 3rd. Cambridge, U.K.: Cambridge University Press, 2020.

[2] A. Vaswani et al. “Attention is all you need”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 30. 2017, pp. 5998–6008.

[3] A. Radford et al. “Robust speech recognition via large-scale weak supervision”. In: *Proc. 40th Int. Conf. Machine Learning (ICML)*. PMLR. 2023, pp. 28492–28518.

[4] A. Baevski et al. “wav2vec 2.0: A framework for self-supervised learning of speech representations”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. 2020, pp. 12449–12460.

[5] A. Babu et al. “XLS-R: Self-supervised cross-lingual speech representation learning at scale”. In: *Proc. Interspeech 2022*. 2022, pp. 2278–2282.

[6] A. Conneau et al. “Unsupervised Cross-lingual Representation Learning at Scale”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. 2020, pp. 8440–8451.

[7] W. Nekoto et al. “Participatory research for low-resourced machine translation: A case study in African languages”. In: *Findings of the Association for Computational Linguistics: EMNLP 2020*. 2020, pp. 2144–2160.

[8] S. H. Imam et al. “Automatic Speech Recognition for African Low-Resource Languages: Challenges and Future Directions”. In: *arXiv preprint arXiv:2505.11690* (2025), pp. 1–8.

[9] Vineel Pratap et al. “Scaling Speech Technology to 1,000+ Languages”. In: *Journal of Machine Learning Research* 25.1 (2024), pp. 1–52.

[10] D. I. Adelani et al. “MasakhaNER 2.0: Africa-centric transfer learning for named entity recognition”. In: *Proc. 2022 Conf. Empirical Methods in Natural Language Processing (EMNLP)*. 2022, pp. 4488–4508.

[11] D. Amodei et al. “Deep Speech 2: End-to-End Speech Recognition in English and Mandarin”. In: *Proceedings of the 33rd International Conference on Machine Learning*. 2016, pp. 173–182.

[12] L. Dong, S. Xu, and B. Xu. “Speech-Transformer: A No-Recurrence Sequence-to-Sequence Model for Speech Recognition”. In: *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2018, pp. 5884–5888.

[13] D. Bigioi and P. Corcoran. “Multilingual video dubbing—a technology review and current challenges”. In: *Frontiers in Signal Processing* 3 (2023), p. 1230755.

[14] J. F. Nunamaker, M. Chen, and T. D. M. Purdin. “Systems development in information systems research”. In: *Journal of Management Information Systems* 7.3 (1990), pp. 89–106.

[15] Harvard Humanitarian Initiative. *Data Collection Tools for Challenging Environments*. `https://www.kobotoolbox.org/`. Accessed: 2026-09-11. 2024.

[16] H.-S. Won et al. “End-to-end multilingual automatic dubbing via duration-based translation with large language models”. In: *Proc. 2025 Conf. Empirical Methods in Natural Language Processing: System Demonstrations*. 2025, pp. 515–521.